6

Graphs

Chapter Learning Outcomes

CLO

6

- Define a graph as a set of vertices and a set of edges.
- Determine the size of a graph.
- Distinguish between directed and undirected graphs.
- Use graphs to model situations in which the vertices represent objects, and edges represent a particular relationship between objects.
- Understand the following graph terminologies: (weighted graph, loop graph, multigraph, pseudograph, regular graph, complete graph, null graph).
- Find the degree of each vertex in directed and undirected graphs.
- Understand the handshaking theorem.
- Distinguish between connected and unconnected graphs.
- Represent the vertices and edges of a graph as an adjacency matrix and an adjacency list.
- Know and understand the bipartite graphs.
- Distinguish between a path and a circuit in a given graph.

Graphs – An Introduction

In many cases we are faced with a problem that is defined in terms of a number of objects or entities that have some relationships among them. The following are examples:

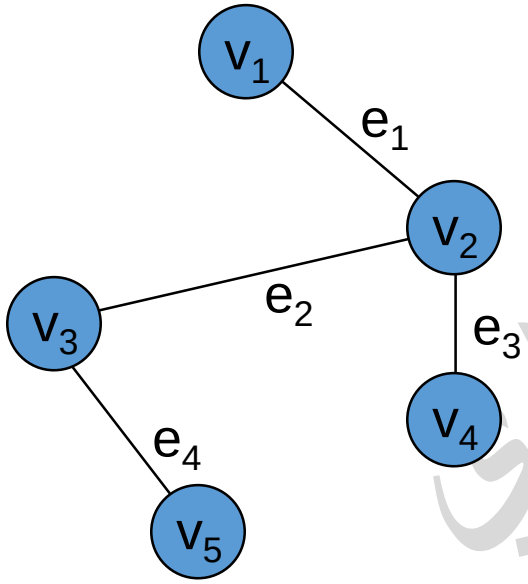
Examples

- Social Networks
- Travelling Salesperson
- Circuit Layout
- Project Scheduling
- Computer Networks
- Flight Scheduling

د. حسن المعيدي

What is a Graph?

A Graph G consists of a set V of vertices or nodes and a set E of edges (arcs) that connect the vertices. We write G=(V,E).



G

Define graph (G)

G=(V,E)

$V=\{v_1,v_2,v_3,v_4,v_5\}$

$E=\{e_1,e_2,e_3,e_4\}$

$e_1=(v_1,v_2)$

$e_2=(v_2,v_3)$

$e_3=(v_2,v_4)$

$e_4=(v_3,v_5)$

What is a Graph?

A graph G is defined as follows:

$$G=(V,E)$$

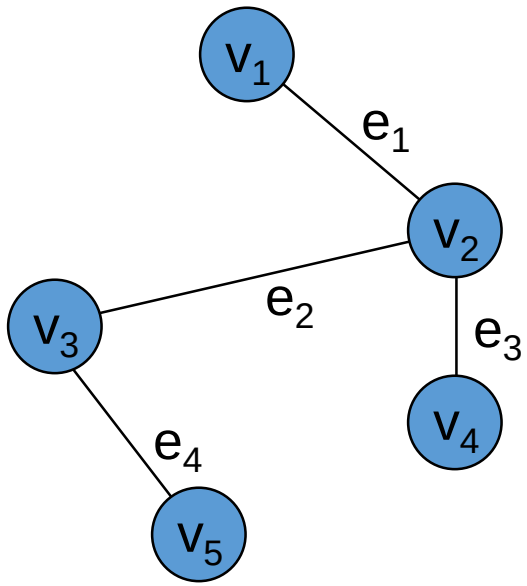
- V(G): a finite, nonempty set of vertices.
- E(G): a set of edges (pairs of vertices).
- An edge is a path between two nodes

د. حسن المعيدي

Size of a Graph

Given Graph G = (V,E) then :

- n = |V| = # vertices
- m = |E| = # edges
- size of G is n + m

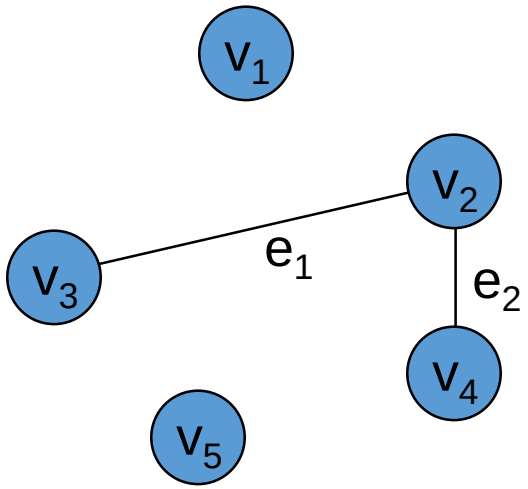


G

د. حسن المعيدي

What is a Graph?

Define the following graphs using formal definition.



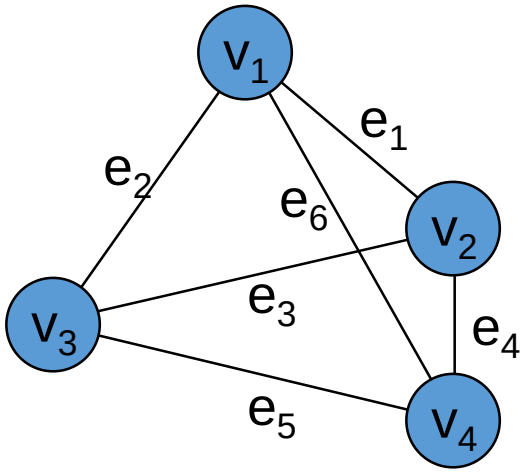
G_1

Define graph (G_1)

د. حسن المعيدي

What is a Graph?

Define the following graphs using formal definition.



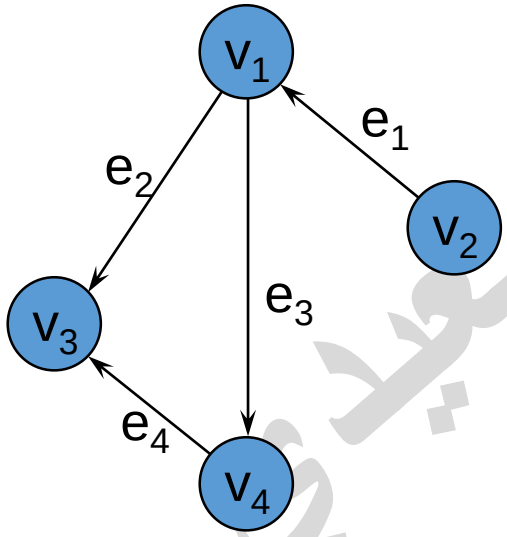
G_2

Define graph (G_2)

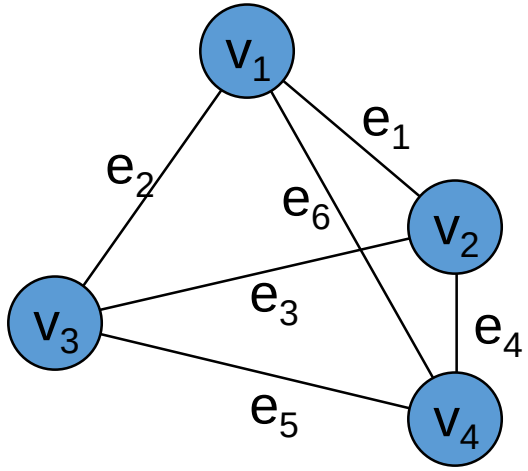
د. حسن المعيدي

Graph Types (Directed & Undirected)

In some cases we want the edges to have directions associated with them; we call such a graph a directed graph or a digraph.



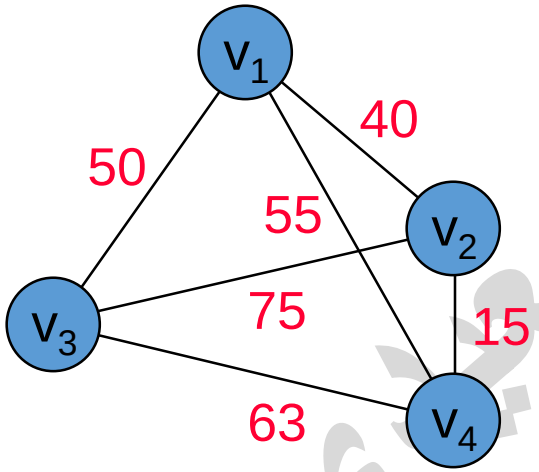
Directed (Digraph)



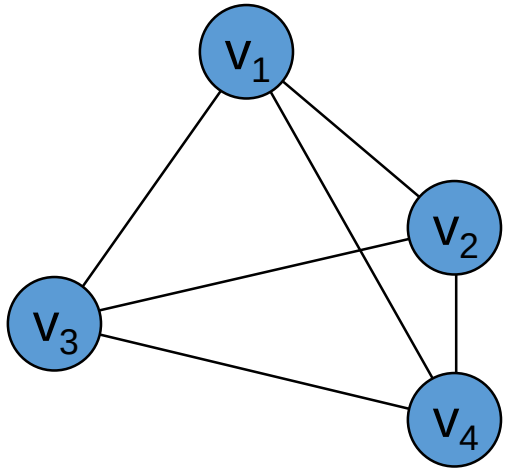
Undirected

Graph Types (Weighted & Unweighted)

In some cases, we want to associate a weight with each edge in the graph. Such a graph is known as a weighted graph.



Weighted Graph

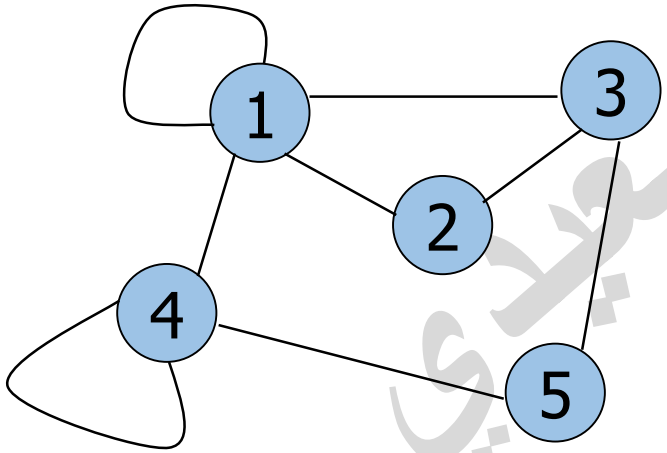


Unweighted Graph

Graph Types (Loop Graph & Multigraph)

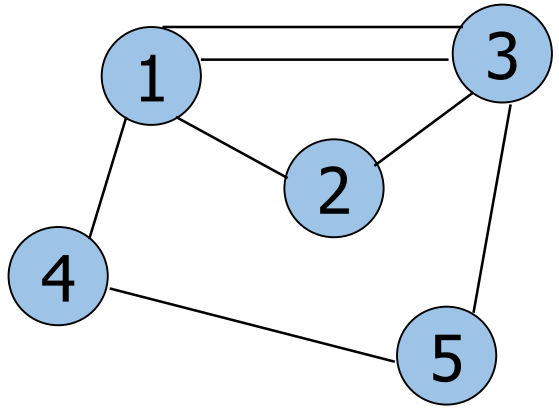
A **loop graph** is a graph that has a **loop**.

a **loop** is an edge that **connects a vertex to itself**



Loop Graph

A **multigraph** is a graph that can have **more than one edge between a pair of vertices.**

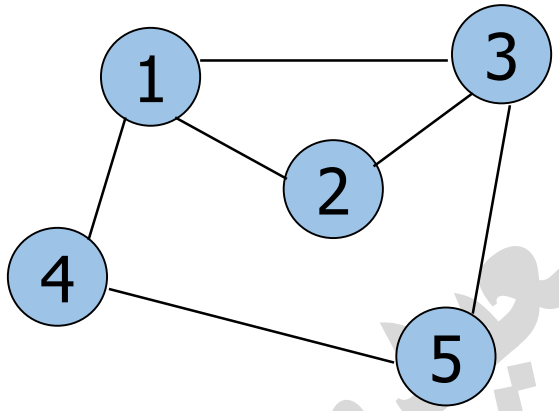


Multigraph

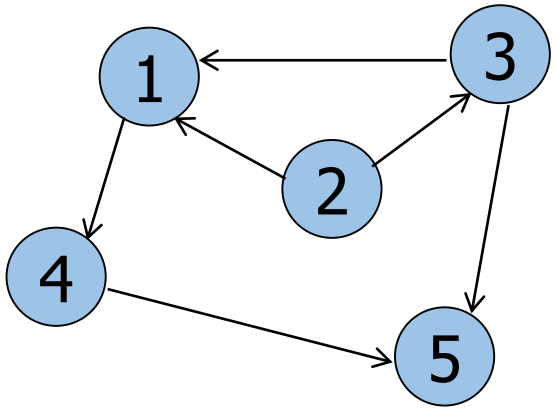
د. حسن المعيدي

Graph Types (Simple Graph)

A **simple graph** (proper graph) has **no loops** and **no multiple edges**.



Simple Graph

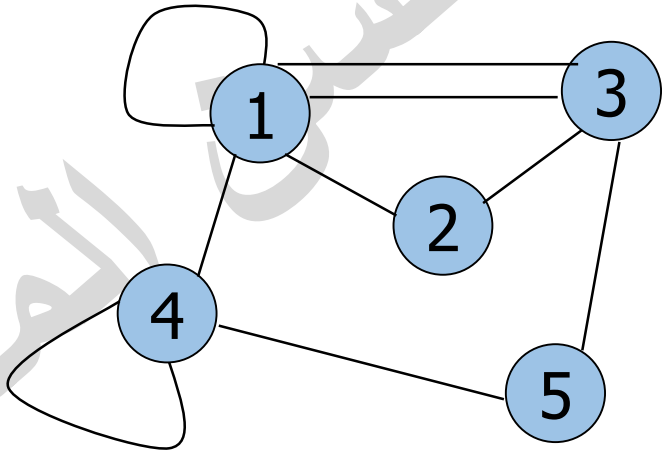


Simple Digraph

د. حسن المعيدي

Graph Types (Pseudograph)

A **pseudograph** may include **loops** as well as **multiple edges**.



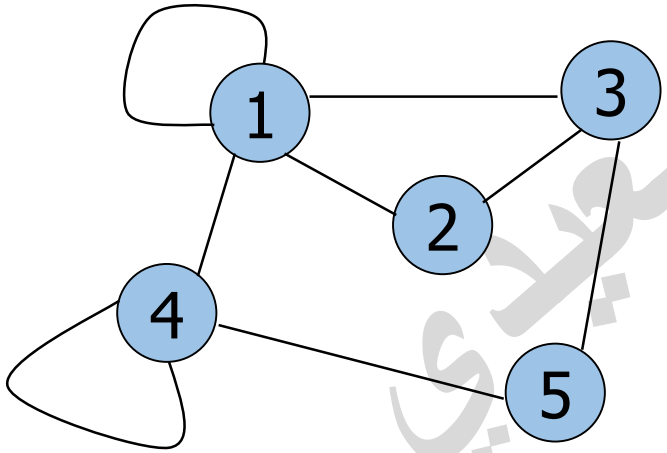
Pseudograph

د. حسن المعيدي

Graph Types (Loop Graph & Multigraph)

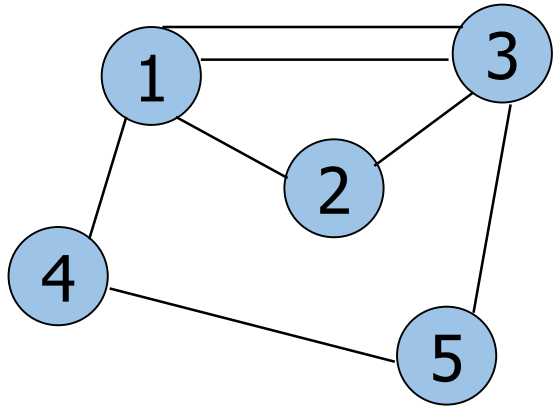
A **loop graph** is a graph that has a **loop**.

a **loop** is an edge that **connects a vertex to itself**



Loop Graph

A **multigraph** is a graph that can have **more than one edge between a pair of vertices.**

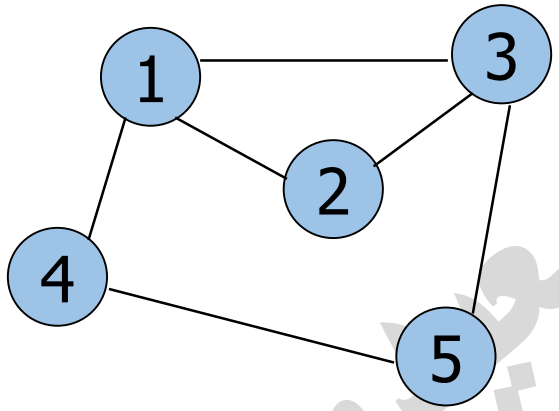


Multigraph

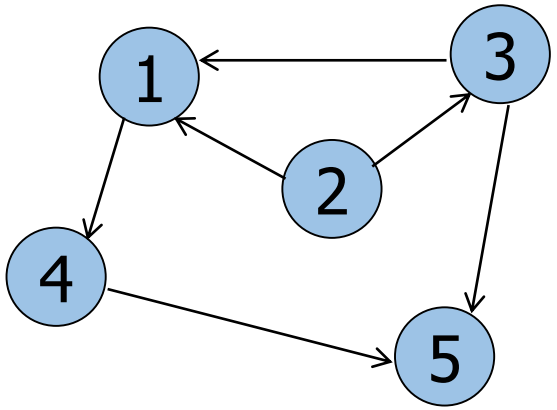
د. حسن المعيدي

Graph Types (Simple Graph)

A **simple graph** (proper graph) has **no loops** and **no multiple edges**.



Simple Graph

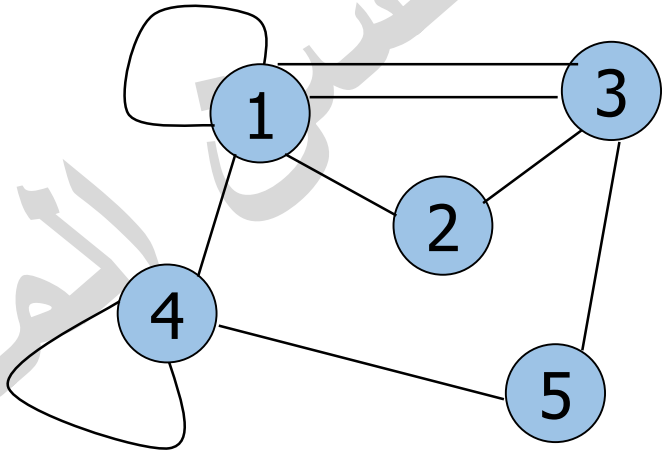


Simple Digraph

د. حسن المعيدي

Graph Types (Pseudograph)

A **pseudograph** may include **loops** as well as **multiple edges**.



Pseudograph

د. حسن المعيدي

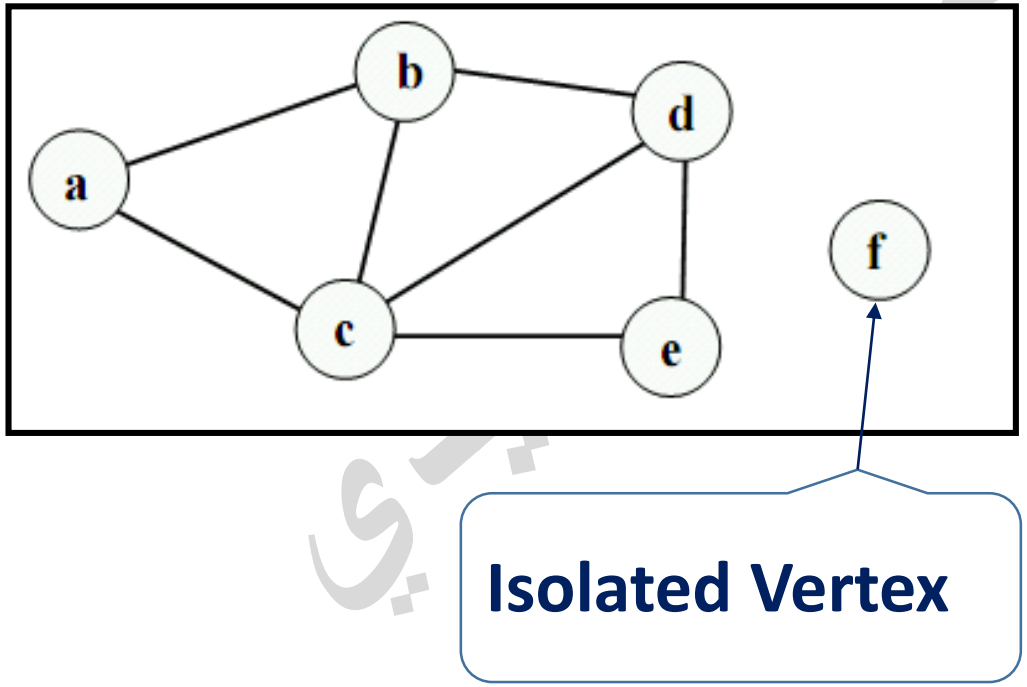
Summary

Term	Edge Type	Multiple Edges Ok?	Loop Ok?
Simple Graph	Undirected	No	No
Loop Graph	Undirected	No	Yes
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple Directed Graph	Directed	No	NO
Directed Loop Graph	Directed	No	Yes
Directed Multigraph	Directed	Yes	Yes
Mixed Graph	Directed & Undirected	Yes	Yes

Undirected Node Degree

Degree in undirected graphs:

Degree(v_i) = # of adjacent (incident) edges to vertex v_i in G.

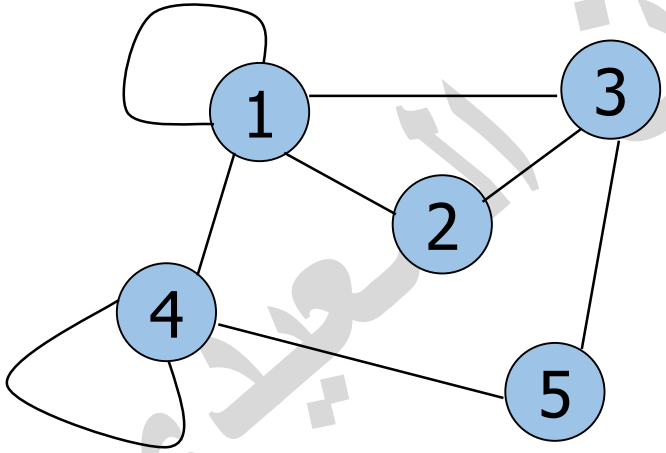


Vertex	Degree
a	
b	
c	
d	
e	
f	

Undirected Node Degree

Example:

Find all the degrees for each vertex in the following graph.



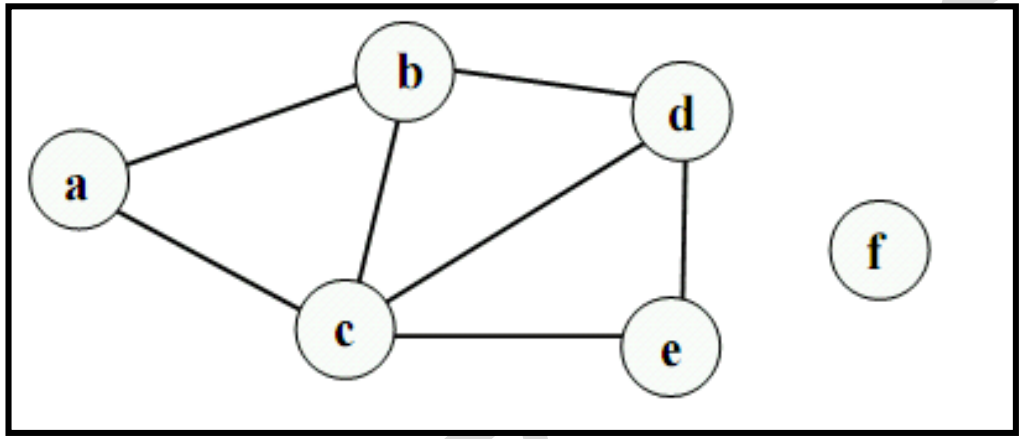
Vertex	Degree
1	5
2	2
3	3
4	4
5	2
Total	16

د. حسن المعيدي

Neighborhoods of Vertices

Degree in undirected graphs:

$N(v_i)$ = Set of all neighbors of a vertex (v_i).

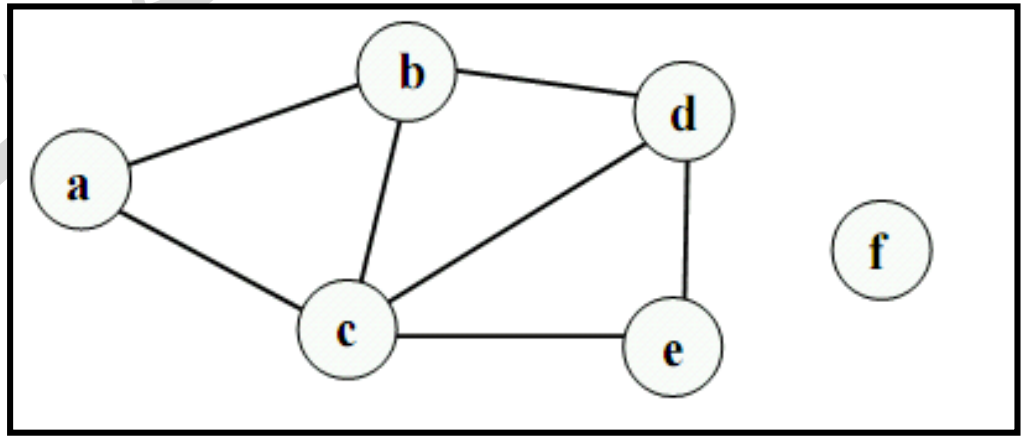


Vertex	Neighborhoods
a	
b	
c	
d	
e	
f	

Handshaking Theorem

(*Handshaking Theorem*): If $G = (V,E)$ is an undirected graph with m edges, then

$$\underbrace{2m = \sum_{v \in V} deg(v)}_{\text{must be even}}$$



Vertex	Degree
a	
b	
c	
d	
e	
f	
Total	

د. حسن المعيدي

Handshaking Theorem (Examples)

EXAMPLE

How many edges are there in an undirected graph with 10 vertices of degree 6?

$$2m = \sum_{v \in V} deg(v)$$

د. حسن المعيدي

Handshaking Theorem (Examples)

EXAMPLE

If an undirected graph has 5 vertices, can each vertex have degree 3?

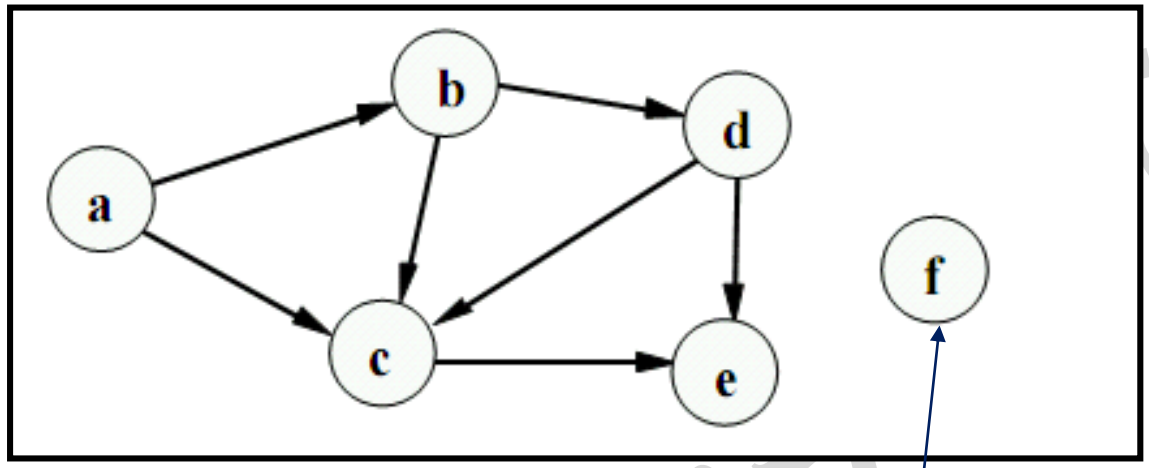
$$2m = \sum_{v \in V} deg(v)$$

د. حسن المعيدي

Directed Node Degree

Degree in directed graphs:

- In-Deg(V) or $Deg^{-}(V)$ = # of incoming edges.
- Out-Deg(V) or $Deg^{+}(V)$ = # of outgoing edges.



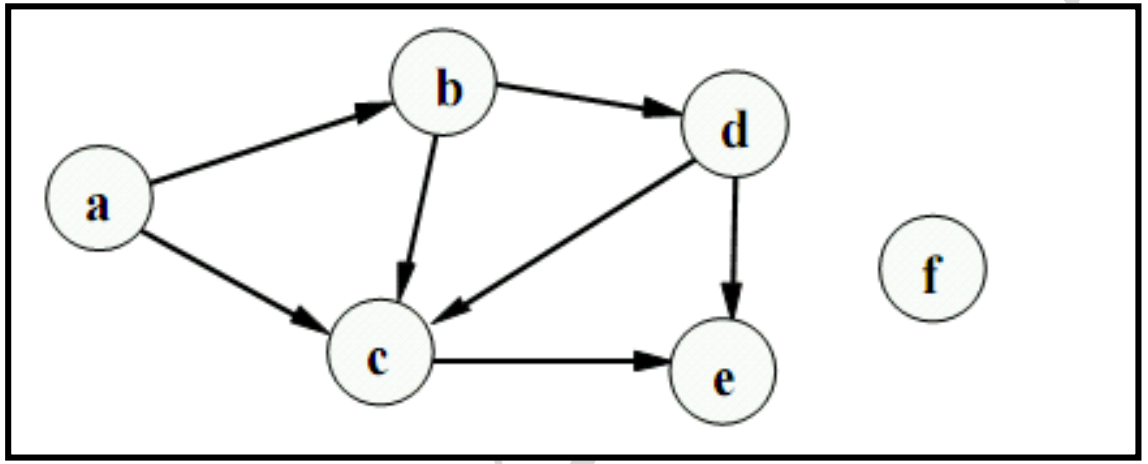
Isolated Vertex

Vertex	In-Degree	Out-Degree
a		
b		
c		
d		
e		
f		

Directed Node Degree

If $G = (V,E)$ is a directed graph with m edges, then

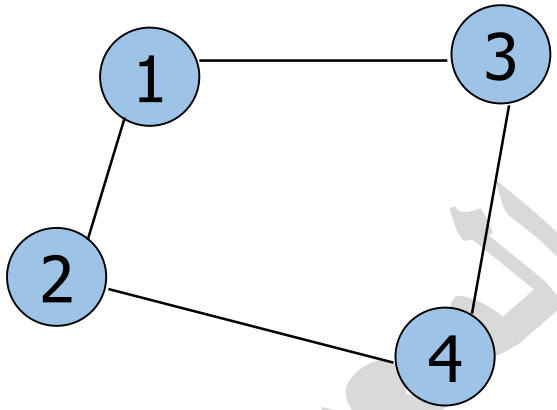
$$m = |E| = \sum_{v \in V} deg^{+}(v) = \sum_{v \in V} deg^{-}(v)$$



Vertex	$Deg^{-}(v)$	$Deg^{+}(v)$
a		
b		
c		
d		
e		
f		
Total		

Graph Types (Regular Graph)

Regular graph (G_R) is an undirected graph where each vertex has the same number of neighbors (every vertex has the same degree).



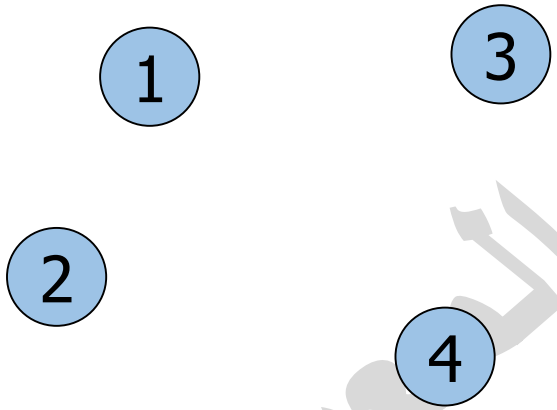
Regular Graph

Vertex	Degree
1	
2	
3	
4	
Total	

د. حسن المعيدي

Graph Types (Null Graph)

Null graph (G_N) is a graph containing no edges



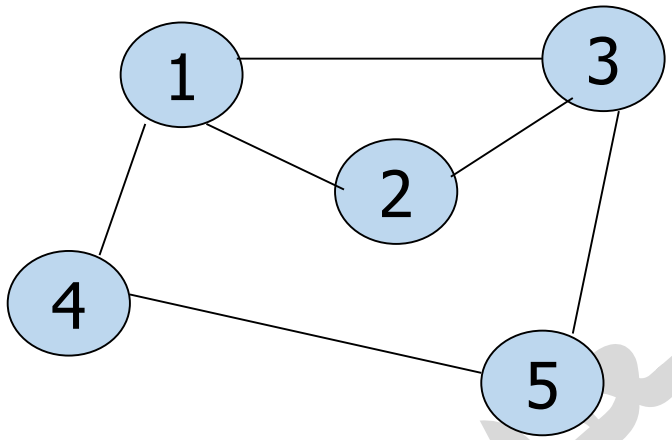
Null Graph

Vertex	Degree
1	
2	
3	
4	
Total	

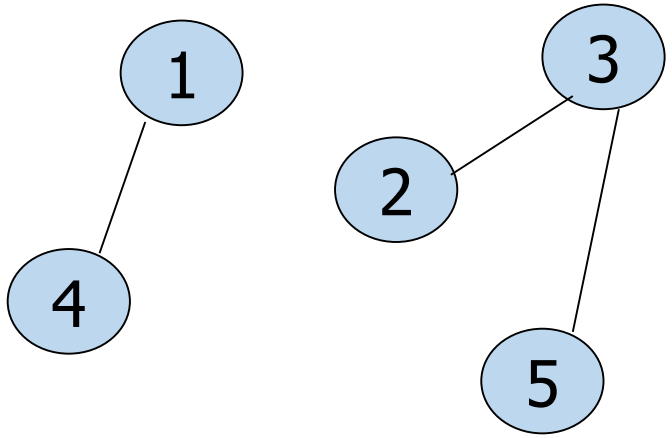
د. حسن المعيدي

Graph Types (Connected Graph)

A graph G is called **connected** if there is at least one path from every vertex to every other vertex in the graph



Connected Graph

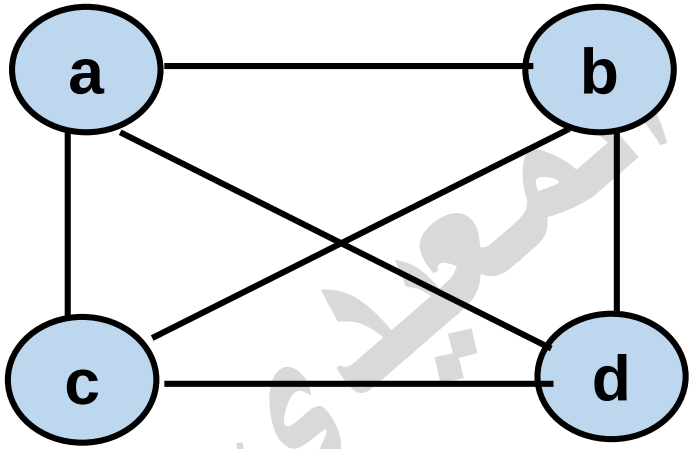


Unconnected Graph

د. حسن المعيدي

Graph Types (Complete Graph)

- A graph G is Completed Graph (G_c) iff every node u in G is adjacent to every other node v in G.
- A completed graph is already connected.
- It has n(n-1)/2 edges.
- The degree for each vertex = (n-1)

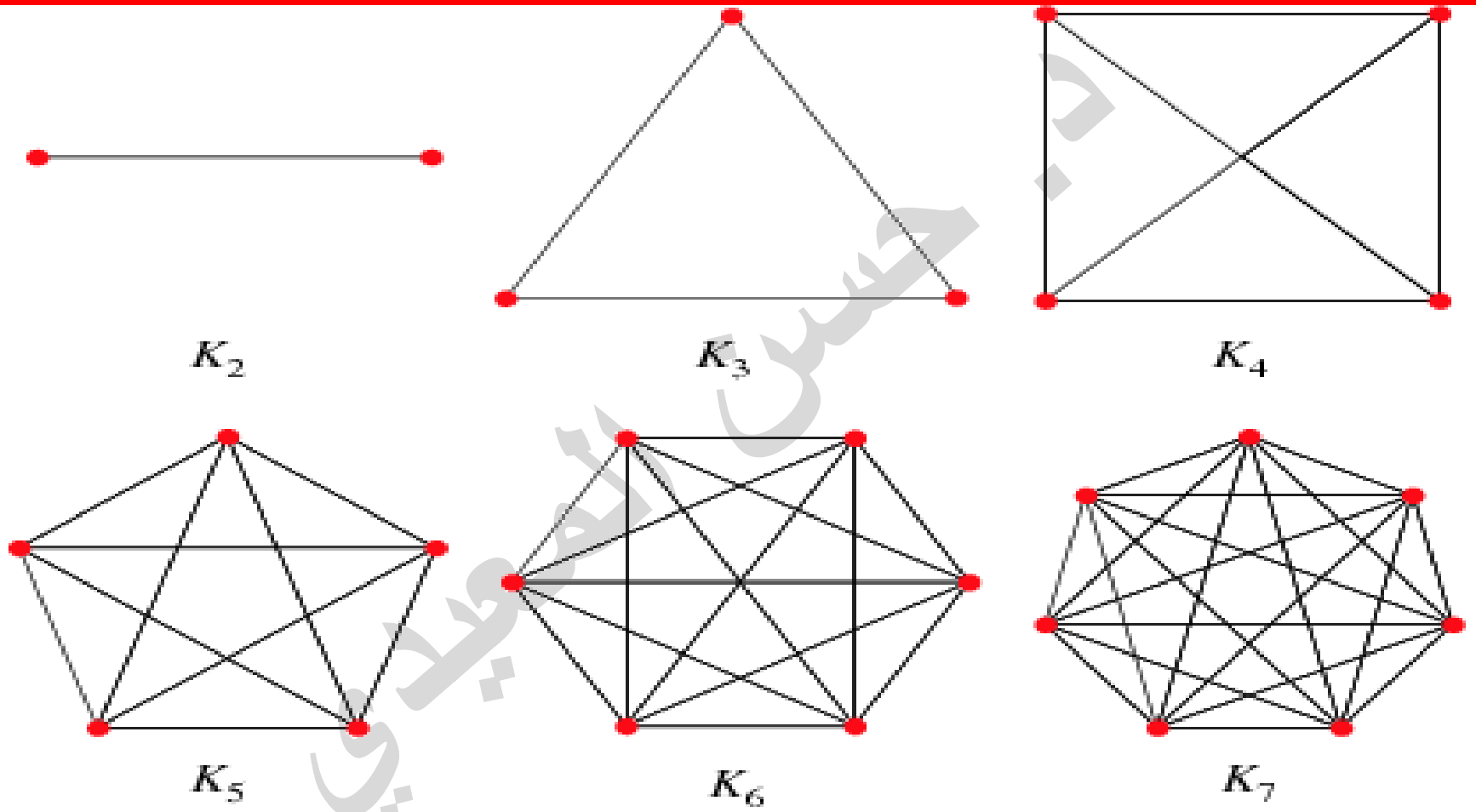


Completed Graph

Vertex	Degree
a	
b	
c	
d	
Total	

د. حسن المعيدي

Graph Types (Complete Graph)



Completed Graphs

د. حسن المعيدي

Graph Representation

There are two ways to represent a graph:

1. Adjacency Matrix (Array-Based Implementation).
2. Adjacency List (Linked-List Implementation).

د. حسن المعيدي

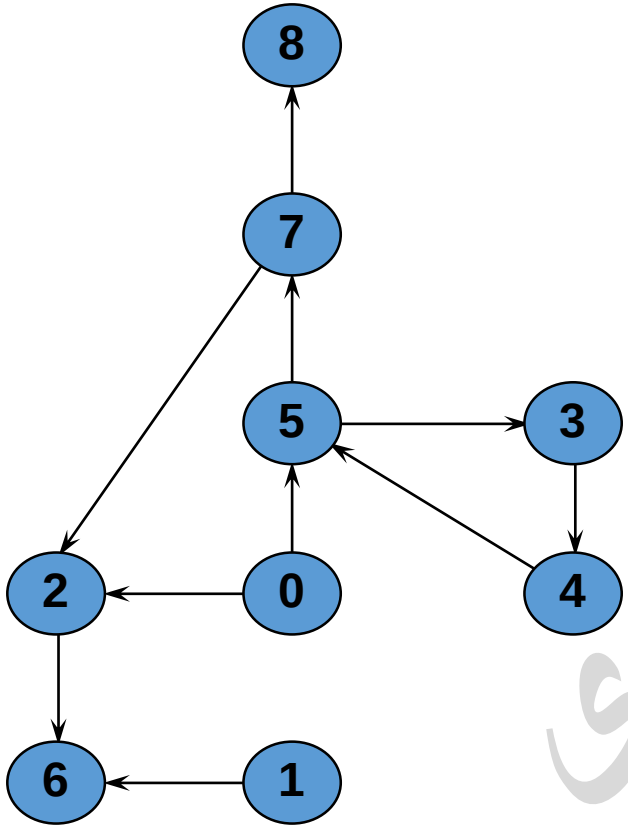
Graph Representation

The adjacency matrix for a graph G=(V,E) with n (or |V|) vertices numbered 0, 1, …, n-1 is an n x n array M such that M[i][j] = 1 if and only if there is an edge from vertex i to vertex j.

The adjacency list for a graph G=(V,E) with n vertices numbered 0, 1, …, n-1 consists of n linked lists. The ith linked list has a node for vertex j if and only if the graph contains an edge from vertex i to vertex j.

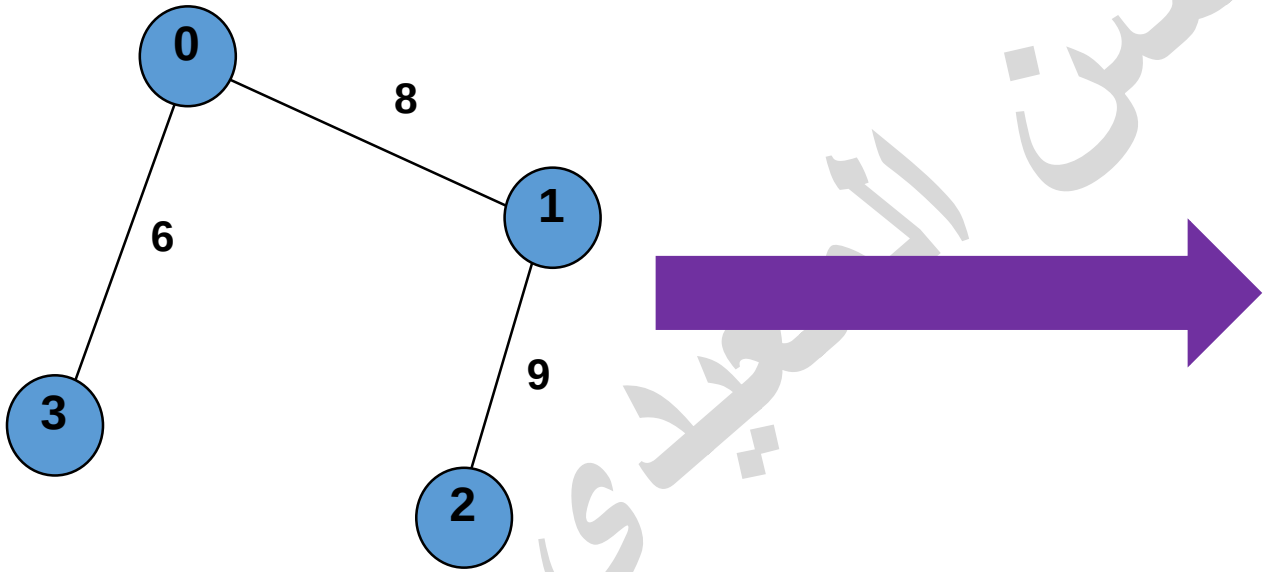
د. حسن المعيدي

Adjacency Matrix



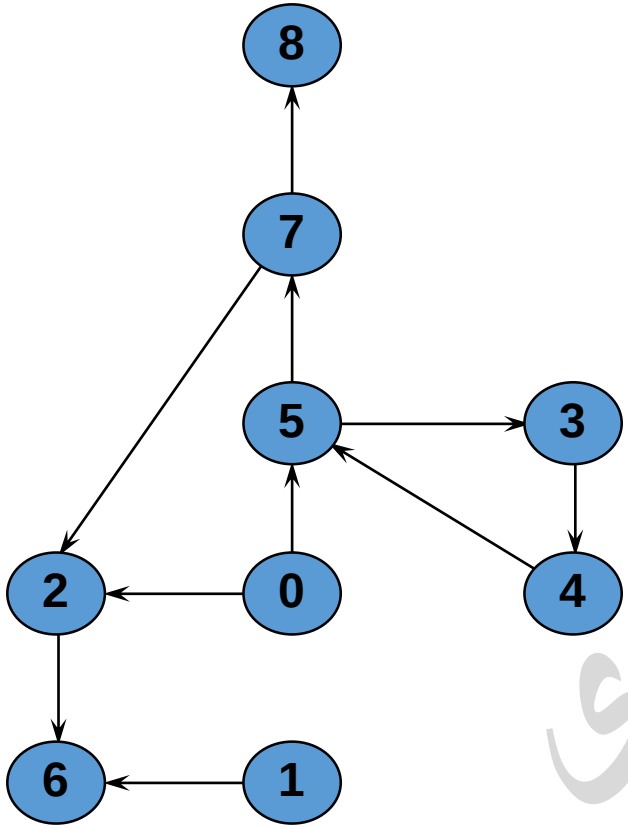
د. حسن المعيدي

Adjacency Matrix



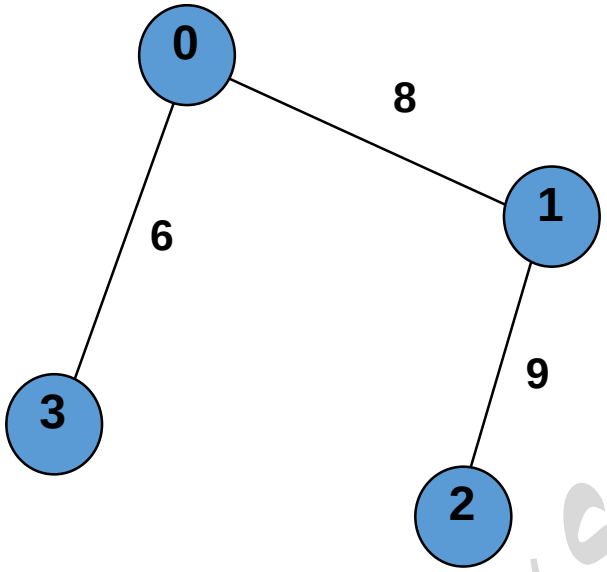
د. حسن المعيدي

Adjacency List



د. حسن المعيدي

Adjacency List



د. حسن المعيدي

Graph Representation

EXAMPLE

If an undirected graph (G) has 5 vertices and 4 edges.

1- How many nodes did you need to represent G using an adjacency matrix?

2- How many nodes did you need to represent G using an adjacency list?

د. حسن المعيدي

Graph Representation (Summary)

Method	Graph Type	Size
Adjacency Matrix	Undirected Graph	
	Directed Graph	

Method	Graph Type	Size
Adjacency List	Undirected Graph	
	Directed Graph	

Bipartite Graphs

Definition: A simple graph G is **bipartite** (K) if V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge connects a vertex in V_1 and a vertex in V_2.

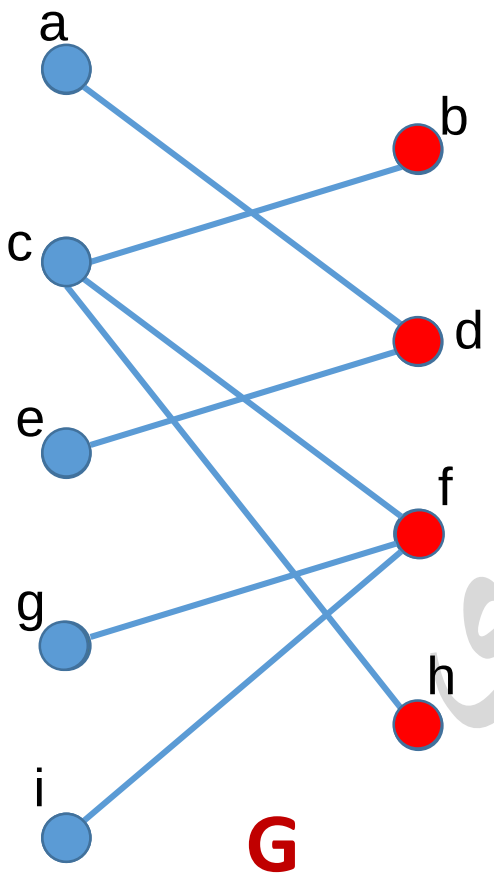
- In other words, there are no edges which connect two vertices in V_1 or in V_2.

- It is not hard to show that an equivalent definition of a bipartite graph is a graph where it is possible **to color the vertices red or blue so that no two adjacent vertices are the same color**.

Bipartite Graphs

EXAMPLE

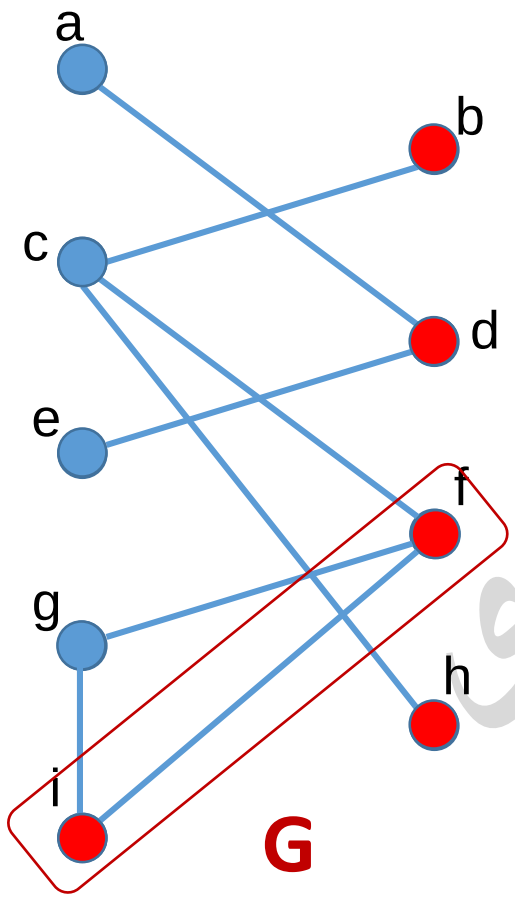
Determine if the following graph G is bipartite or not?



د. حسن المعيدي

Bipartite Graphs

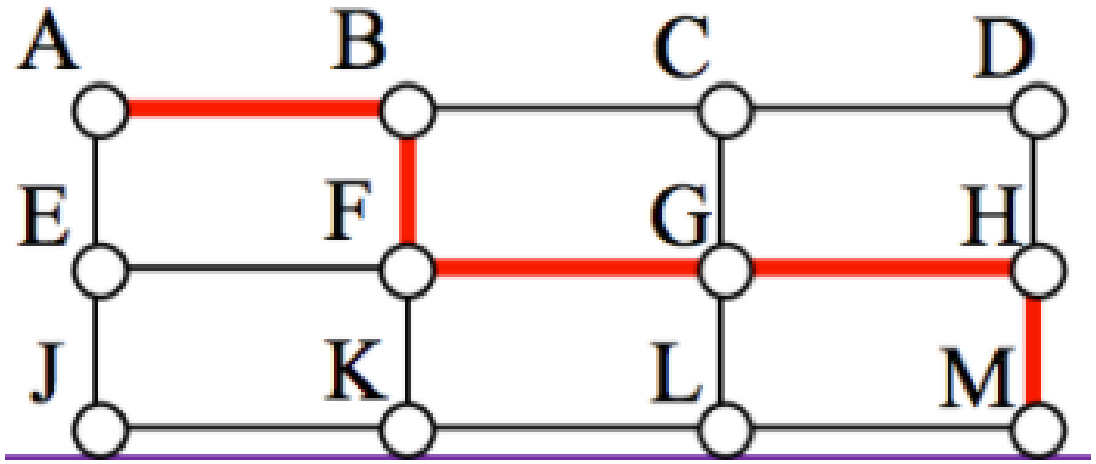
Example: **Determine if the following graph G is bipartite or not?**



د. حسن المعيدي

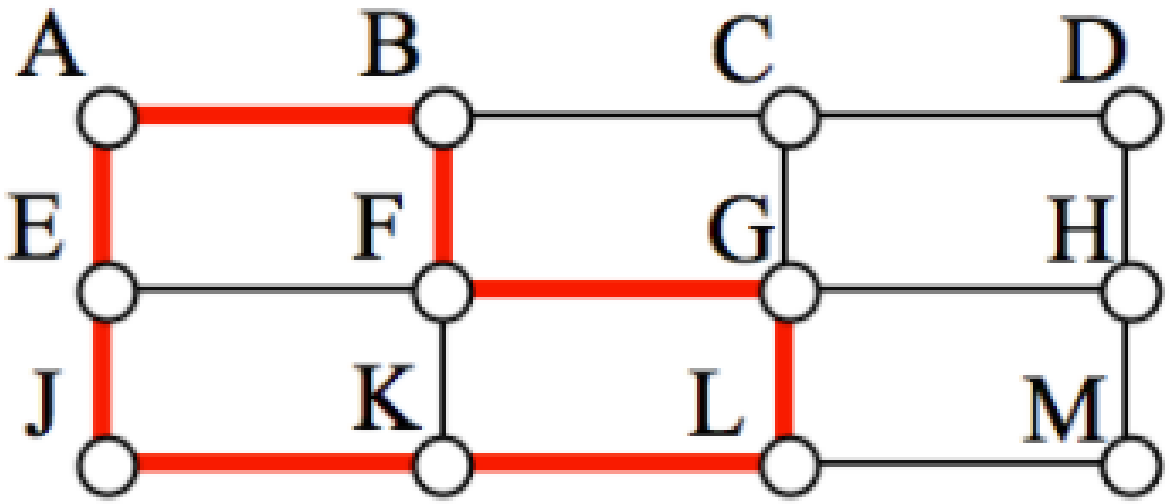
Path

A **path** is a sequence of vertices using the edges. Usually we are interested in a path between two vertices. For example, a path from vertex A to vertex M is shown below. It is one of many possible paths in this graph.



Circuit

A **circuit** is a path that begins and ends at the same vertex. A circuit starting and ending at vertex A is shown below.



A **path or circuit is simple** if it does not contain the same edge more than once.

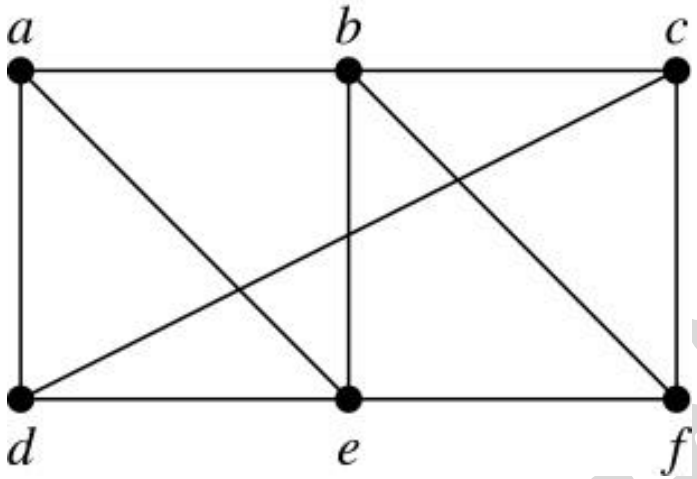
Paths

Definition: Let n be a nonnegative integer and G an undirected graph. A *path* of *length n* from u to v in G is a sequence of n edges $e_1, \ldots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \ldots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \ldots, n$, the endpoints x_{i-1} and x_i.

- When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \ldots, x_n$(since listing the vertices uniquely determines the path).
- The path is a ***circuit*** if it begins and ends at the same vertex ($u = v$) and has length greater than zero.
- A **path or circuit is *simple*** if it does not contain the same edge more than once.

Paths

Example:

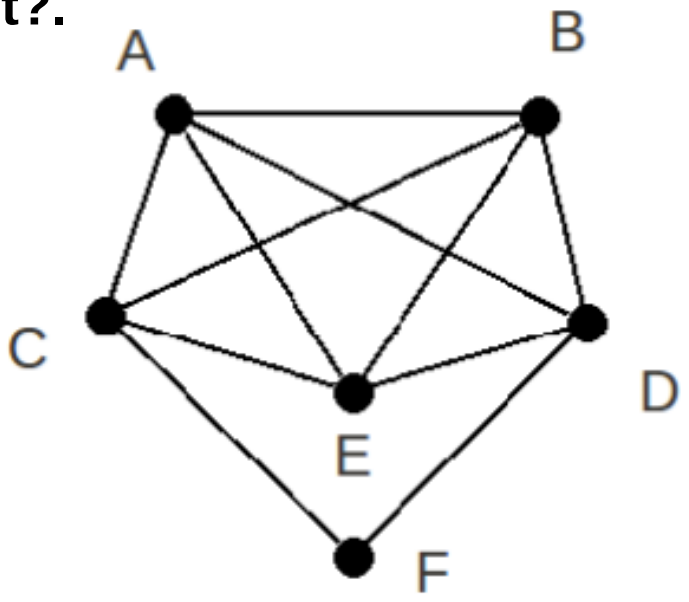


- *a, d, c, f, e* is a simple path of length 4.
- *d, e, c, a is not a path because e is not connected to c.*
- *b, c, f, e, b is a circuit of length 4.*
- *a, b, e, d, a, b is a path of length 5, but it is not a simple path.*

د. حسن المعيدي

Review Question

1. Determine if the following graph is bipartite or not?.
2. What is the type of this graph?.
3. Find the degree of all vertices.
4. Are E and B adjacent vertices?.
5. Are E and F adjacent vertices?.
6. Find a path of length 4 for F to C.
7. Find a circuit of length 4 for B to B.



د. حسن المعيدي

Check - Chapter Learning Outcomes

CLO

6

- Define a graph as a set of vertices and a set of edges.
- Determine the size of a graph.
- Distinguish between directed and undirected graphs.
- Use graphs to model situations in which the vertices represent objects, and edges represent a particular relationship between objects.
- Understand the following graph terminologies: (weighted graph, loop graph, multigraph, pseudograph, regular graph, complete graph, null graph).
- Find the degree of each vertex in directed and undirected graphs.
- Understand the handshaking theorem.
- Distinguish between connected and unconnected graphs.
- Represent the vertices and edges of a graph as an adjacency matrix and an adjacency list.
- Know and understand the bipartite graphs.
- Distinguish between a path and a circuit in a given graph.

End of Chapter

6